

each transaction of their token. It creates a truly shared environment, where network participants are rewarded at all levels for the value they add to the ecosystem. Hydra also offers DEX (digital employee experience) capability.

The buzz around HydraChain is due to its unique features. Its inflation/deflation system ensures that the chain is secured with market driven mechanisms, and the fixed transaction fees allow seamless operation.

The emergence of HydraChain

In 2021, the HydraChain quickly grew from a simple network into a strong community-based economy. It has robust infrastructure and top tier node coverage across the globe. A diverse set of projects is being built on it.

Hydra is becoming a truly decentralised blockchain thanks to more than 900 nodes scattered across the globe. A healthy geographical distribution across six continents not

only adds repetitiveness to the network, but also helps it protect against local risks and country-specific challenges.

There are several infrastructures built by HydraChain which help develop a network. Some of these include Hydra DEX, Hydra cross-chain bridge with Ethereum, mobile wallet app, browser extension wallet, web wallet, Hydra Explorer and staking client.

HydraChain as power saver

A single Bitcoin transaction as of 2022 releases 1,207kg of CO₂. The total annual carbon footprint of Bitcoin is 114 megatons of CO₂, which is equivalent to the Czech Republic's nationwide energy consumption for a full year!

With Hydra, POS (point of sale) terminals can be run on Raspberry Pi and even on free tiers of AWS. A Raspberry is capable of mining Hydra blocks with as little as 5W of power consumption. Every block mined via Hydra and not Bitcoin approximately saves 2.5T of CO₂ as carbon waste.

Some key characteristics of Hydra are:

- Decentralised governance protocol
- Dynamic nature of network settings and hybrid operations between UTXO and EVM layers
- Fixed transaction fees set in USD equivalent
- Burn on transaction fees
- Gas royalty for smart contract creators
- Flexible supply management via inflationary and deflationary elements
- A true proof-of-stake consensus that enables every single user to stake without any requirement for a minimum number of coins
- An average output of 75-85 transactions per second (TPS) with an elastic burst scalability of up to 540TPS
- One-click installers for running a node on an average household computer
- Various technical optimisations and repetitions

Factor	HydraChain	Bitcoin	Ethereum
Energy consumption/ year (3 nodes/year)	0.00052 TW	77.78 TW	14.81 TW
Transaction fees	0.17 ADA (\$0.21)	.00039 BTC (\$20)	0.012 ETH (\$20)
Scalability	Supported by core consensus protocol	Slow adoption of lightning network	No ETA
Formal verification	Available	Not available	Not available
Tx conformation time	60 seconds	1800 seconds	39 seconds
Smart contract support	Available. Written in languages like Solidity or Serpent	Not available	Available. Written in languages like Solidity, Vyper, etc
NFTs (non-fungible tokens)	Available	Not available	Available
Block limit	Has limitations in creation of blocks. It creates new blocks only if there is a pending transaction	Block limit of 1MB	No block limit
Algorithm	Uses a unique consensus algorithm called Hydra Consensus	Runs on SHA-256 hash algorithm	Runs on Ethcash proof-of-work algorithm
Inception year	2021	2009	2015
Founder/Creator	Nikola Alexandrov	Satoshi Nakamoto	Vitalik Buterin
Security	Secured transaction	Robust security and immutability	Less security

Table 1: Comparison between HydraChain, Bitcoin and Ethereum (Table credit: forum.cardano.org)